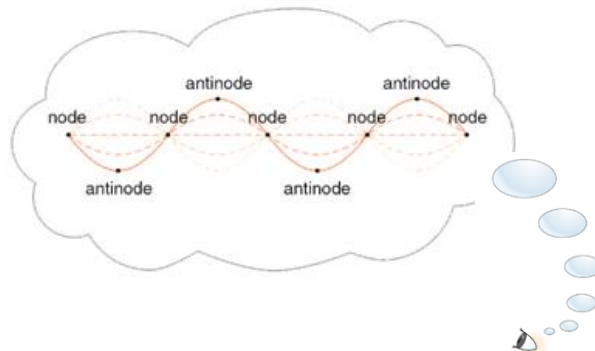
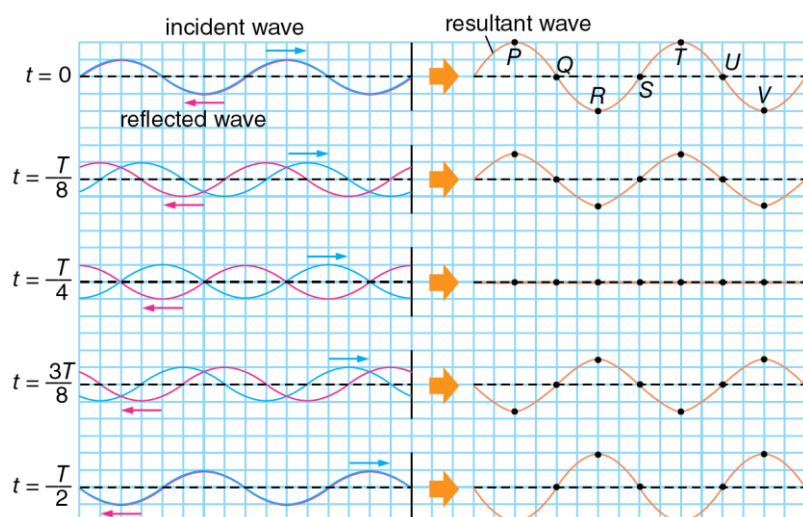
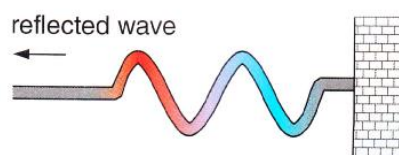
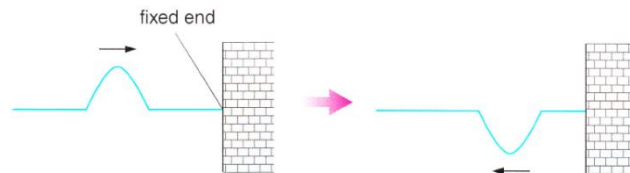


Stationary waves

No phase change



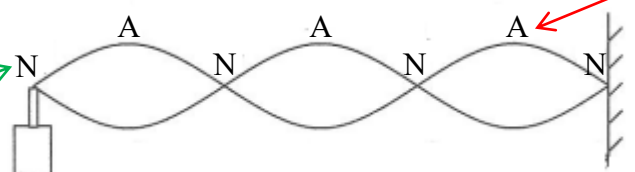
Phase change



A **stationary wave** is formed by the **superposition** of two waves with the **same frequency** and **amplitude** travelling at the **same speed** in **opposite directions**

Characteristics of stationary wave

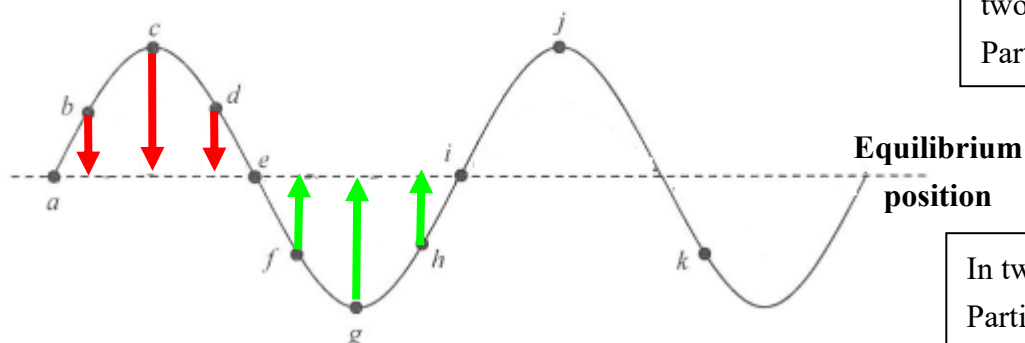
Positions on a stationary wave where particles **do not vibrate** are called **nodes**.



Positions where the particles vibrate with **maximum amplitudes** are called **antinodes**.

- Amplitude of the stationary wave formed is **twice** that of the original wave.
- Different particles vibrate with **different amplitudes** in a stationary wave.
- The distance between 2 adjacent nodes or 2 adjacent antinodes is equal to half of the wavelength.
- The number of loops would change due to a change of speed or frequency.

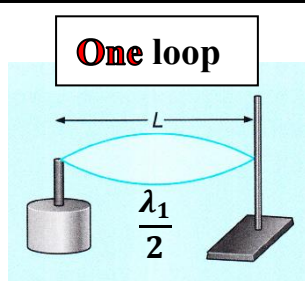
Phase relationship in stationary wave



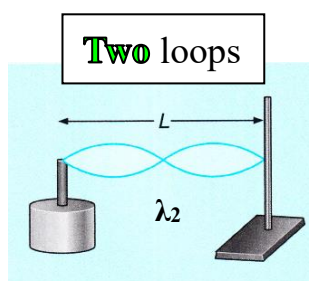
In the **same loop** (between two adjacent nodes)
Particles vibrate **in phase**

In two adjacent loops
Particles vibrate in **antiphase**

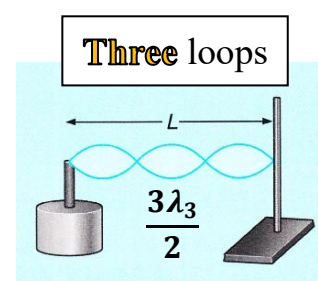
Stationary wave on a string



One loop
Fundamental frequency (f_1)

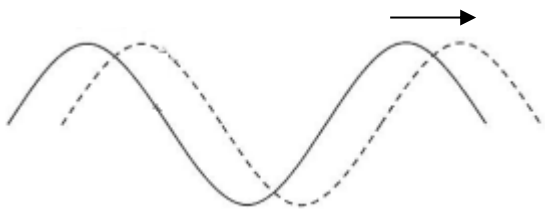
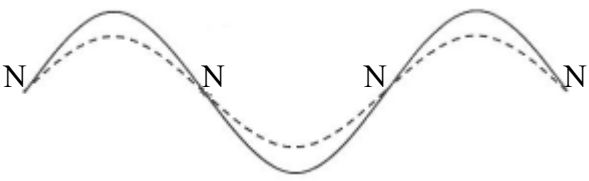


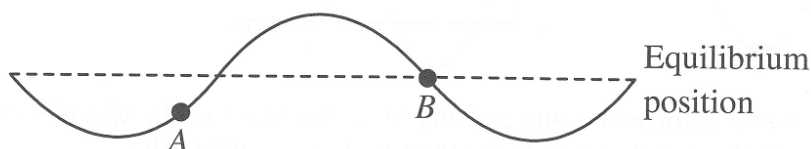
$f_2 = 2f_1$



$f_3 = 3f_1$

Comparison between travelling and stationary

Travelling wave	Stationary wave
	
A travelling wave moves forwards .	A stationary wave does not move from one place to another.
All particles vibrate with the same amplitude .	Different particles vibrate with different amplitudes .
Different particles reach their maximum displacements at different time .	All particles reach their maximum displacements at the same time .
Neighbouring particles always vibrate out of phase	Particles in the same loop vibrate in phase . Particles in two adjacent loops vibrate in antiphase .
Energy is transmitted from one place to another through wave motion.	Energy is localized .



1. The waveform of a wave is shown.

(a) Mark the point that is in phase with point A if the wave is a

(i) stationary wave; or

(1 mark)

(ii) progressive wave.

(1 mark)

(b) Describe the motion of B if the wave is a

(i) stationary wave; or

(1 mark)

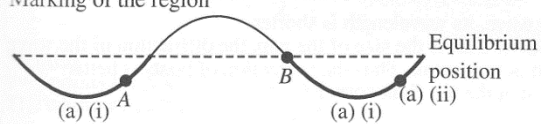
(ii) progressive wave.

(1 mark)

(c) Compare the wavelength and frequency of the wave if the wave is a stationary wave with those of the wave if the wave is a progressive wave.

(2 marks)

(a) (i) Marking of point
(ii) Marking of the region

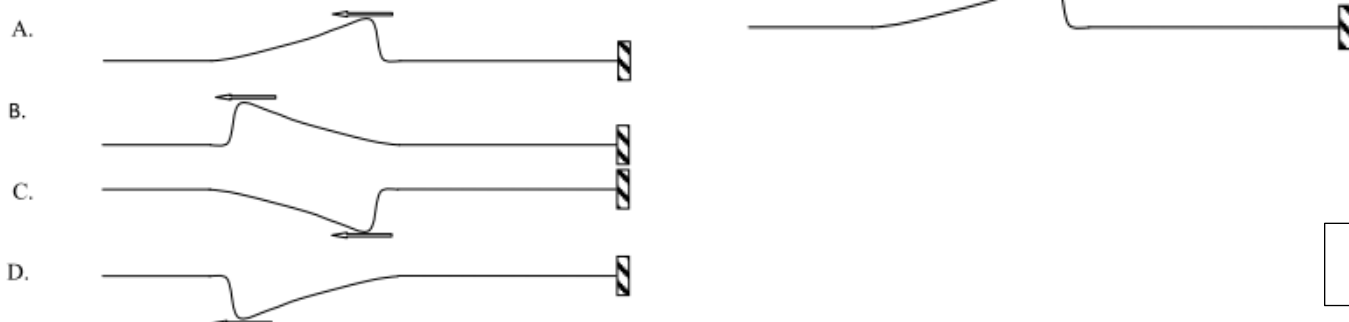


(b) (i) It is always at rest.
(ii) It moves with maximum speed.

(c) Both waves have the same wavelength and frequency.

2. 2012 DSE MC Q.16

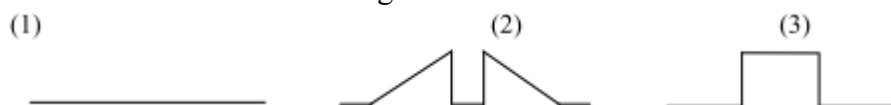
A pulse on a string propagates towards the right end which is fixed. Which of the following represents the reflected pulse?



D

3. 2015 DSE MC Q.14

Two pulses of the same shape travel along a stretched string with one end fixed to the wall as shown above. Which of the following can be the resultant waveform at different instants later?

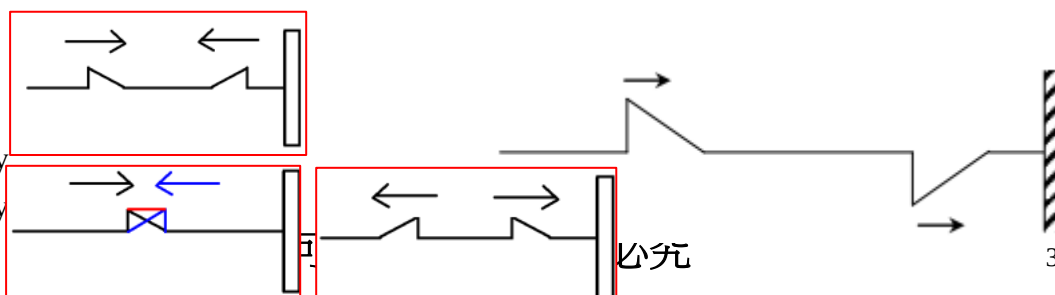


A. (1) only

B. (3) only

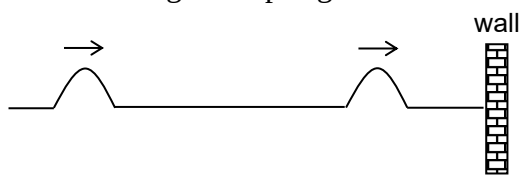
C. (1) and (2) only

D. (2) and (3) only



D

4. Two identical transverse pulses are travelling on a spring towards an end fixed to a wall as shown below.



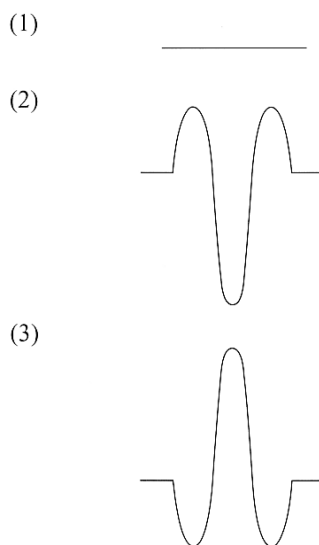
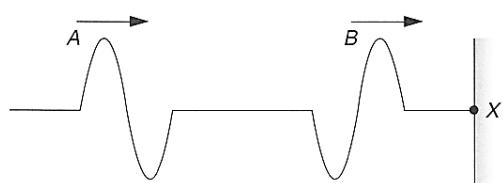
Which of the following shape of the spring at a particular moment is/are possible? (Assume that the amplitudes of the pulses do not change during their movement.)



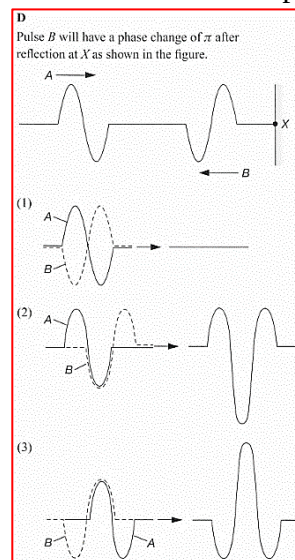
- A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

D

5. Two pulses A and B of equal amplitude travel towards a fixed end X along a stretched as shown in the figure. Which of the following can be resultant wave pattern at different instants when the two pulses meet?



- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

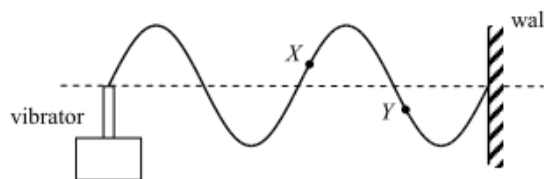


D

6. 2016 DSE MC Q.18

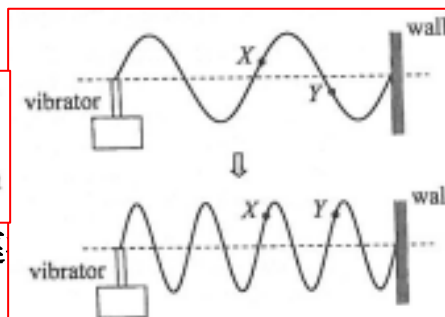
A string is tied to a vibrator while the other end is fixed to a wall. A stationary wave is formed as shown. Which statement is correct when the frequency of the vibrator doubles?

- A. The wavelength will double
B. The wave speed will double
C. The amplitude will be halved
D. Particles X and Y become vibrating in phase



D

- A * Since the wave speed does not depend on the frequency of the vibrator, it will remain unchanged. By $v = f\lambda$, the wavelength will be halved.
B * (See the explanation in A.)
C * Since the amplitude does not depend on the frequency of the vibrator, it will remain unchanged.

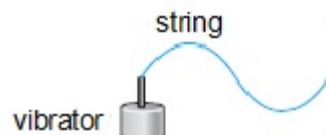


警告: 不可翻印, 販賣, 違

7. The experimental set-up shown below can be used to produce stationary wave patterns on an elastic string by adjusting the frequency f of the vibrator. Which of the following statements are correct?

- (1) Stationary wave patterns can be observed for more than one value of f .
- (2) When f increases, the number of nodes in the stationary wave pattern formed increases.
- (3) When f increases, the speed of the waves on the string increases.

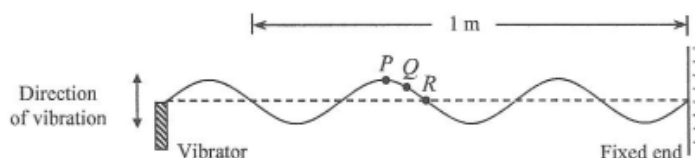
- A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)



A

8. A vibrator generates a stationary wave on a string. The above diagram shows the string at the instant of maximum displacement. Which of the following statements is **incorrect**?

- A. Particles P and Q move with the same amplitude.
- B. The motions of particles P and Q are in phase.
- C. Particle Q is momentarily at rest at this instant.
- D. The wavelength of the stationary wave is 0.4 m.



A

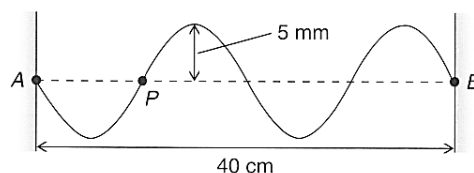
- * A. At this instant, P is the antinode which has the maximum amplitude. Q has smaller amplitude.
- ✓ B. Particles in the same loop must move in the same phase
- ✓ C. Q is at maximum displacement and must be momentarily at rest
- ✓ D. Since distance between 2 nodes is half of the wavelength which is 0.2 m $\therefore \lambda = 0.4$ m

9. A stationary wave is produced along a string between two fixed points A and B. The figure shows the waveform off the stationary wave at certain instant. Which of the following can be deduced from the figure?

- (1) Point P is a node.
- (2) The amplitude of the stationary wave is 5 mm
- (3) The wavelength of the stationary wave is 20 cm

- A (1) only
B (2) only

- C (1) and (3) only
D (2) and (3) only



C

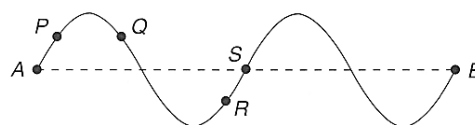
- C
- (1) Point P is always at rest.
 - (2) The amplitude could be larger than 5 mm.
 - (3) Wavelength $\lambda = 40 \div 2 = 20$ cm

10. A stationary wave is formed on an elastic string with two fixed ends A and B. The figure shows the waveform of the stationary wave at a certain instant where particle P is moving upwards. Which of the following statements is/are correct?

- (1) Point Q is moving downwards
- (2) Particles P and R are always in anti-phase
- (3) Particle S is a node of the stationary wave

- A (1) only
B (2) only

- C (1) and (3) only
D (2) and (3) only



D

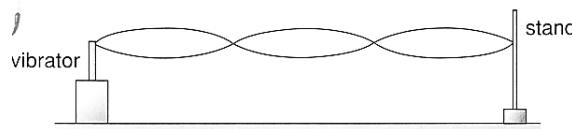
- D
- (1) Particles on the same envelope are always in phase.

11. Different stationary waves can be formed on an elastic string by adjusting the frequency f of the vibrator as shown in the figure. Which of the following statements is/are correct?

- (1) When f increases, the speed of the waves on the string increases
(2) When f increases, the number of nodes on the string increases
(3) The sound wave produced by the vibration of the string has the same frequency as the stationary wave on the string

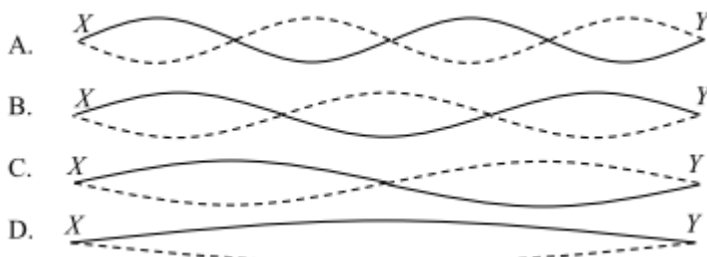
- A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

D
(1) Wave speed depends on the medium only.
(2) When f increases, the wavelength λ decreases, so the number of nodes increases.



D

12. A string is set to vibrate at frequency f such that a standing wave is formed between two fixed supports X and Y as shown. If the tension in the string is increased by adding weight gradually while the frequency is kept at f , which of the following is a possible mode of vibration at a steady state?



D

Speed of transverse waves along stretched string or spring: $c = \sqrt{\frac{T}{\mu}}$ where

T: Tension in the string, μ : Mass per unit length (For reference only)

As the tension in the string is increased, speed of the transverse wave along the string increases.

Since the frequency is unchanged, by $v = f\lambda$, the wavelength increases.

The only option that shows an increase of wavelength of standing wave is D.

13. A spring with one end fixed to the wall is stretched to 5 m long. The other end of the spring is vibrated to produce stationary wave patterns. The minimum frequency of vibration to produce a stationary wave pattern on the spring is 4 Hz. What is the speed of waves on the spring?

- A 20 m s⁻¹
B 40 m s⁻¹
C 60 m s⁻¹
D 80 m s⁻¹

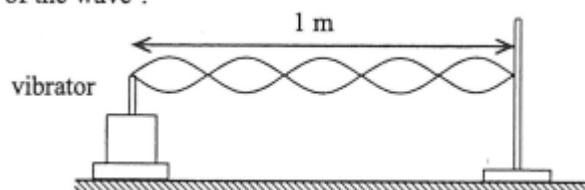
B

14. 2018 DSE MC.18

The figure shows a string with one end fixed and the other end tied to a vibrator. A stationary wave is formed as shown at a certain frequency.

If the speed of the wave along the string is 7 m s⁻¹, what is the frequency of the wave?

- A. 2.8 Hz
B. 7 Hz
C. 17.5 Hz
D. 35 Hz



C

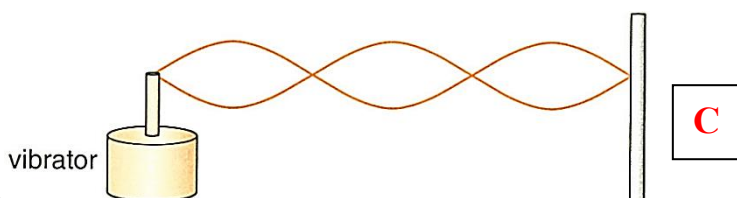
15. A stationary wave is produced by a vibrator on a string as shown. The frequency of the vibrator is 420 Hz. To produce a stationary wave of 5 loops on the string, what should be the frequency of the vibrator?

- A. 140 Hz
B. 252 Hz
C. 700 Hz
D. 2100 Hz

By $f_n = nf_1$, •.....

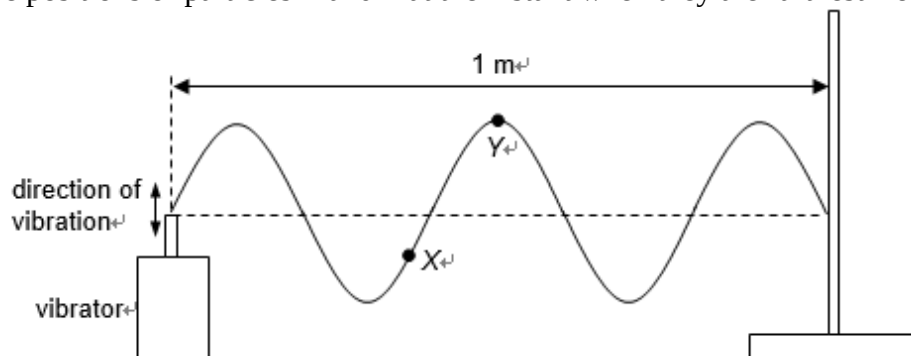
$$f_1 = \frac{f_n}{n} = \frac{420}{3} = 140 \text{ Hz}$$

$$f_5 = 5 \times f_1 = 5 \times 140 = 700 \text{ Hz}$$

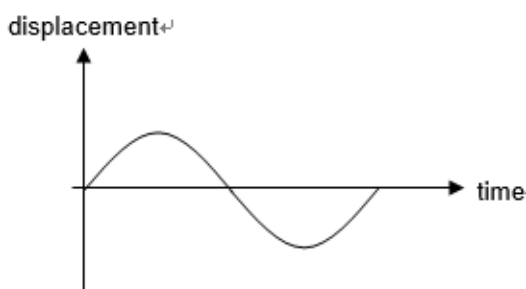


C

16. A stationary wave is produced on an elastic string which is fixed with one end to a vibrator and the other to a stand. The frequency of the vibrator is set at 50 Hz. The distance between the vibrator and the stand is 1 m. The figure below shows the positions of particles X and Y at the instant when they are furthest from their equilibrium positions.



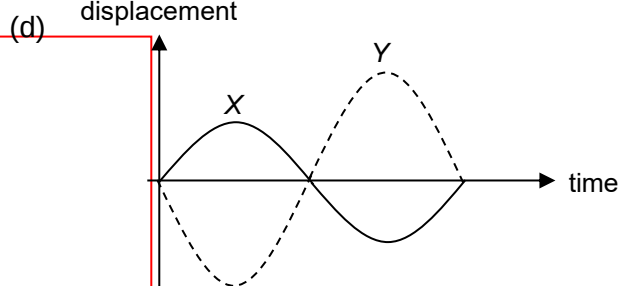
- Find the wavelength and the wave speed of the stationary wave. (3 marks)
- In the figure, label an antinode with A and a node with N. (2 marks)
- Describe the motion of particles X and Y at the instant shown. (2 marks)
- The figure below shows the displacement–time graph of particle X.



Sketch the displacement–time graph of particle Y in the same figure. (2 marks)

- Describe how you can produce another stationary wave with fewer antinodes compared with the one shown.

- They are momentarily at rest.

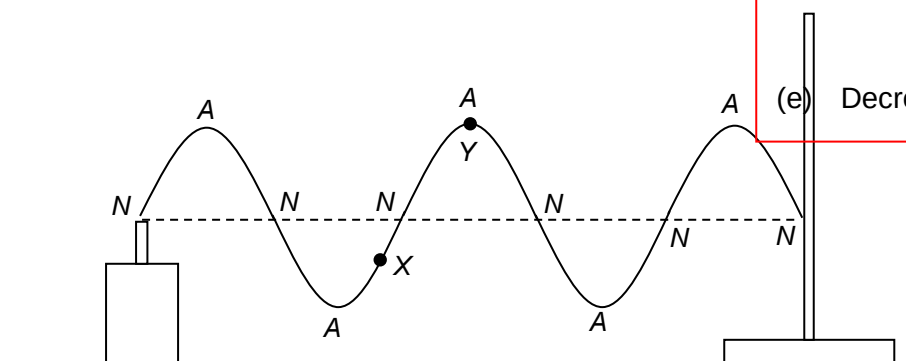


- Decrease the frequency of the vibrator gradually.

(a) Wavelength = $\frac{1}{2.5} = 0.4 \text{ m}$

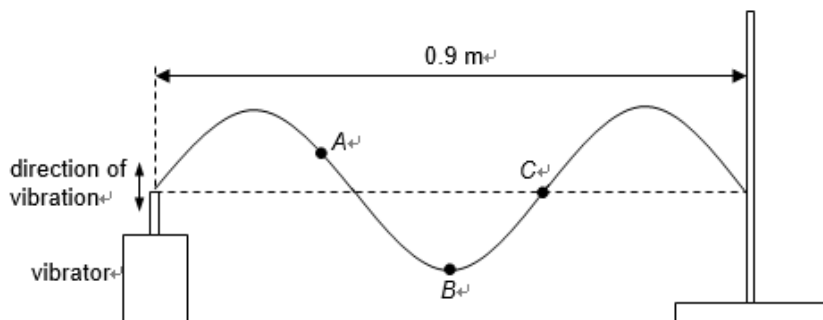
Wave speed = $f\lambda$
 $= 50 \times 0.4$
 $= 20 \text{ m s}^{-1}$

(b)



17. A stationary wave is generated on a string as shown. The vibrator, 0.9 m away from the stand, is vibrating at 25 Hz. At the instant shown, the string is at the maximum displacement.

- Find the wavelength of the stationary wave. (1 mark)
- Compare the oscillation of particles A, B and C in terms of amplitude and phase relationship. (3 marks)
- How long does it take for particle A to complete one oscillation? (2 marks)
- Sketch the shape of the string after 0.01 s. Mark the positions of particles A, B and C on the string. (4 marks)

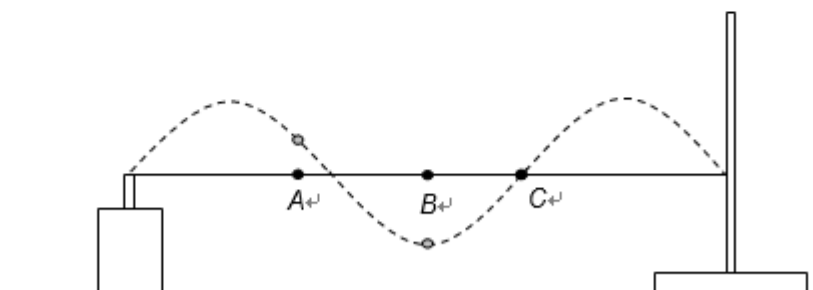


(a) Wavelength = $\frac{0.9}{1.5} = 0.6 \text{ m}$

- (b) Particle A vibrates with a smaller amplitude than particle B.
Particle C does not vibrate.
Particles A and B vibrate in antiphase.

(c) Time for one oscillation = $\frac{1}{f}$
 $= \frac{1}{25}$
 $= 0.04 \text{ s}$

(d)



(Correct shape of string)

(Correct positions of particles)